

H2GUARD · DESIGN DECISION RECORD

H2Guard TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	HGD-DDR-001	Version	0.1
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/h2guard		
Source	docs/decisions/0001-trl2-review-decisions.md @ ebb853d		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted for TRL 3 work under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. The recommendations for items D1 to D9 are adopted for TRL 3 work pending Amish's review; items O1 to O5 remain proposed, awaiting Amish.

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed eight items as "Proposed, awaiting Amish", and the design precis HGD-PRC-001 v0.2 listed eight key design choices, each with options and a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed this batch item by item. Every item that carries a recommendation is therefore adopted as recommended for TRL 3 work, open for his review; items without a recommendation, and the budget figure, stay open.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in HGD-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Items adopted for TRL 3 work.

#	Item	Adopted choice	Status
D1	Sensor baseline	Option (a): catalytic sensor for the trip plus a metal oxide sensor for early warning (about \$50 of sensors), with a certified molecular property spectrometer sensor documented as the upgrade for any room used without close supervision	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D2	Set points	Warning at 10 % LFL (0.4 % vol), trip at 25 % LFL (1.0 % vol)	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D3	Supply valve	Normally closed, energize to open, rather than a latching or motorized valve	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D4	Ventilation	Continuous exhaust with a boost, rather than a fan that starts on alarm	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D5	Inventory rule (R9)	Hydrogen that could be released into the room is 1 % of room volume or less at atmospheric pressure, or a flow restrictor limits release to the design leak; a condition for H2Bench and any other lab project using H2Guard. H2Bench is asked, through the review note, to reflect it	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D6	Reset and coverage	Key-switch reset; one detector head per room as the base kit, with a second head as an option for beamed or pocketed ceilings	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D7	First users	A university teaching lab first, because supervision and a technician are in place	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D8	Independent hardware trip	A comparator trip beside the microcontroller; kept as the core of the design	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D9	Supply voltage	24 V DC throughout, from a certified plug-in supply	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review

Table 2. Items that remain open.

#	Item	Status
O1	Budget. The TRL 2 review recommended raising <code>budget_usd</code> from \$180 to \$250. <code>budget_usd</code> is not changed. HGD-CAL-001 puts the parts at \$264, over both figures, so the review note asks Amish to choose again.	Proposed, awaiting Amish
O2	Oxygen depletion monitoring when inert gas cylinders share the room. No recommendation was made.	Proposed, awaiting Amish
O3	Where the set points and the inventory rule come from in each country the first users are in. No recommendation was made.	Proposed, awaiting Amish
O4	How airflow is proven: tachometer only, or a differential pressure switch. No recommendation was made; HGD-CAL-001 Table 4 shows a blocked duct is not detected with the tachometer alone.	Proposed, awaiting Amish
O5	Low-cost solenoid valve or a certified gas valve, pending the hydrogen compatibility and seat leak rating. No recommendation was made.	Proposed, awaiting Amish

■ Consequences

- `project.yaml`: only the TRL fields change. `budget_usd` stays at \$180 (O1). No pitch or problem rewording was recommended, so none was applied.
- HGD-PRB-001, HGD-PRC-001 and HGD-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed"; they are adopted for TRL 3 work pending Amish's review. R9 becomes an adopted installation rule. No requirement target is relaxed or redefined by these items.
- The TRL 3 calculations (HGD-CAL-001) led to three changes within the adopted choices: a larger mixed-flow fan so the boost is reached against the duct losses (D4), thinner flame arrestor discs with the sensors close behind them, and a bump test cup, tube and port for R12. They raise the parts cost from \$239 to \$264.
- New proposals from HGD-CAL-001 (a timed escalation from warning to trip, a head placement rule, a redefinition of R13, a revised budget figure) are listed in `docs/REVIEW.md` and await Amish; this record does not adopt them.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.